Emerging Non-Hermitian Topology in a Chiral-Driven-Dissipative Bose-Hubbard Model

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We introduce a driven-dissipative Bose-Hubbard chain describing coupled lossy photonic modes, in which time-reversal symmetry is broken by a coherent drive with a uniform phase gradient. We investigate this model by means of a Gaussian variational ansatz and numerically prove that the steady-state solution is stabilized by an inhomogeneous profile of the driving amplitude, which damps out boundary effects. Our calculations unveil a nonequilibrium phase diagram showing low- and high-density phases for photons separated by a phase coexistence region in which the system exhibits the phenomenon of topological amplification and is characterized by a finite non-Hermitian winding number. Our Letter shows the emergence of non-Hermitian topological phases in an interacting model that can be naturally implemented with superconducting circuits.

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Introduction—The study of topology in condensed matter physics has unveiled a plethora of exciting phenomena like the quantum Hall effect [1] and topological insulators with time-reversal symmetry [2,3]. Topological order in fermionic systems can also appear in the presence of dissipation [4]. In the field of topological photonics [5–8], light propagation is controlled by topological effects leading to applications such as photon routing [9] and directional amplifiers [10,11]. Photonic setups are inherently open quantum systems subjected to photon loss and gain processes, similarly to other bosonic systems like polaritons [12], and vibrational [13] or phononic lattices [13,14]. Topology in bosonic systems is often studied at the single-particle or semiclassical level, but less is known about the interplay between topology, interactions, and dissipation.

Driven-dissipative systems [15,16] are ideal candidates for the study of such an interplay and, in particular, Bose-Hubbard driven-dissipative lattices [17,18] have been extensively investigated. This model consists of nonlinear photonic modes coupled by hopping terms and subject to both photon loss and coherent driving by an external field. They have been addressed by theoretical methods such as the coherent-state ansatz [19,20]; the Gutzwiller ansatz [21]; the truncated Wigner approximation [22]; Keldysh path-integral techniques [23,24]; and the corner renormalization group method [25], showing a nonequilibrium phase diagram with a first-order phase transition between high- and low-density phases as a function of the driving strength. More recent works have explored critical slowing down [22], entanglement [26], quantum chaos [27], and pattern formation [20]. Driven-dissipative Bose-Hubbard chains and lattices can be naturally implemented in the quantum limit of small photon occupations using superconducting circuits [28,29], where strong-interaction effects [30], and even a dissipative quantum phase transition, have been experimentally observed [31]. Other implementations are microcavity polaritons [32], trapped ions [33,34], and ultracold atoms [35–37] and molecules [38].

A natural path to induce topological phases in quantum driven-dissipative systems is to break the time-reversal symmetry. This approach has been recently pursued with the study of incoherently or parametrically driven linear photonic lattices [11,39–42], leading to the discovery of topological amplification. This phenomenon is intimately linked to non-Hermitian topology, since a non-Hermitian dynamical matrix governs linear dissipative systems [43,44]; it is related to so-called point-gap topology [42,45]; and it manifests itself as a directional amplification of input signals.

In this Letter, we study a dissipative Bose-Hubbard chain in which time-reversal symmetry is broken by adding a site-dependent phase to the coherent drive. We explore the emergence of non-Hermitian topological amplification phases via the interplay between interactions and dissipation. Our theoretical description is based on a Gaussian variational ansatz to approximate the density matrix [46,47]. Our main results are (i) stable steady-state solutions can be achieved by using an inhomogeneous profile for the coherent drive that smoothens boundary effects; (ii) the nonequilibrium phase diagram is divided into low- and high-density phases separated by a phase coexistence region—in the latter, the system exhibits non-Hermitian topology and topological amplification properties; (iii) within the phase coexistence region, fluctuations build up at the frontier between topological and trivial phases; and (iv) correlations between sites are enhanced at the topological phase.

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Driven-dissipative Bose-Hubbard model—We consider N nonlinear photonic modes forming a chain and described by annihilation (creation) operators a_j ($a_j^\dagger$), whose unitary dynamics is governed by the Hamiltonian

$$H_a = \sum_{j=1}^{N} \Delta_j a_j^\dagger a_j + J \sum_{j=1}^{N-1} \left(a_{j+1}^\dagger a_j + a_j^\dagger a_{j+1}\right) + U \sum_{j=1}^{N} (a_j^\dagger a_j)^2 + \sum_{j=1}^{N} \epsilon_j \left(a_j e^{i\psi_j} + a_j^\dagger e^{-i\psi_j}\right), \quad (1)$$

where Δ_j are local mode energies, J is a uniform photonic hopping rate, ϕ is a constant hopping complex phase, and U is a local nonlinearity. A coherent drive acts over the array with local amplitude ϵ_j and site-dependent phase ψ_j. The latter is assumed to depend linearly on the site position $\psi_j = j\phi$. The gauge transformation $a_j \to a_j e^{-i\psi_j}$ amounts to the substitution $a_{j+1}^\dagger a_j + a_j^\dagger a_{j+1} \to a_{j+1}^\dagger a_j e^{i\phi} + a_j^\dagger a_{j+1} e^{-i\phi}$, such that the constant coherent drive gradient results in a synthetic gauge field for photons. In the following, we will work in this gauge. Note that this term is analogous to the constant phase acquired by electrons in the presence of a magnetic field after the Peierls substitution, whereas an electric field would result on a time-dependent phase.

The whole quantum dynamics of the chain is determined by a master equation which also includes photon decay:

$$\mathcal{L}(\rho) = -i[H_a, \rho] + \frac{\kappa}{2} \sum_j \left(2a_j \rho a_j^\dagger - a_j^\dagger a_j \rho - \rho a_j^\dagger a_j\right). \quad (2)$$

Photon decay is an essential process in our model, which otherwise would fail to produce a stable steady state. Hereafter, we shall mostly focus on steady-state properties, i.e., on $\rho^\star$ such that $\mathcal{L}(\rho^\star) = 0$. The model in Eqs. (1) and (2) has been investigated theoretically in [27] for a ring of $N = 3$ cavities. Here, we show that an open chain with $N \gg 1$ leads to an entirely new phenomenology related to topology.

Gaussian ansatz—Integrating the quantum evolution described by Eq. (2) is challenging. However, in the limit of weak nonlinearities, a Gaussian ansatz is a good approximation to the exact steady state of the system [46,48]. We start by defining time-dependent mean values $\alpha_j(t) = \langle a_j(t) \rangle$ and fluctuation operators $b_j = a_j - \alpha_j(t)$ fulfilling $\langle b_j \rangle = 0$. Specifically, we write equations of motion for α_j and correlators $G_{jk} = \langle b_j^\dagger b_k \rangle$ and $F_{jk} = \langle b_j b_k \rangle$, fully specifying a Gaussian state. This is a convenient basis for direct application of Wick's theorem on correlators of the form $\langle B_1 B_2 B_3 B_4 \rangle = \langle B_1 B_2 \rangle \langle B_3 B_4 \rangle + \langle B_1 B_3 \rangle \langle B_2 B_4 \rangle + \langle B_1 B_4 \rangle \langle B_2 B_3 \rangle$, with $B_j = b_j$ or $b_j^\dagger$. The ansatz has some desirable properties that have been proven in the case of closed quantum systems; namely, Gaussian states form a so-called Kähler manifold, which implies that equations of motion can be derived consistently from variational principles [47,48]. The master equation in Eq. (2) can be viewed as the system's reduced-state evolution resulting from the joint unitary dynamics of the system plus bath. Applying a Gaussian ansatz to such a joint dynamics corresponds exactly to our approximation, since the reduced state of a Gaussian state is also Gaussian. Hence, in the case of a Markovian bath, the Gaussian ansatz applied to dissipative dynamics enjoys the same properties as those for unitary dynamics (see Supplemental Material [49] for details).

Applying Wick's theorem to the Heisenberg equations of motion, one gets

$$\begin{aligned}
\frac{d\alpha_j}{dt} &= -i\left[\epsilon_j + (\Delta_j + U)\alpha_j + 2U(|\alpha_j|^2 \alpha_j + 2\alpha_j G_{jj} + \alpha_j^* F_{jj}) + J(\alpha_{j+1} e^{-i\phi} + \alpha_{j-1} e^{i\phi})\right] - \frac{\kappa}{2}\alpha_j, \\
\frac{dG_{jk}}{dt} &= iU\left(2(\alpha_j^*)^2 F_{jk} + 4|\alpha_j|^2 G_{jk} + 2F_{jk} F_{jj}^* + 4G_{jj} G_{jk}\right) + i\Delta_j G_{jk} + iJ(G_{j-1k} - G_{jk+1}) e^{-i\phi} - \frac{\kappa}{2} G_{jk} + (j \leftrightarrow k)^*, \\
\frac{dF_{jk}}{dt} &= -iU\left(4|\alpha_k|^2 F_{jk} + 2\alpha_j^2 G_{jk} + 4G_{jj} F_{jk} + 2G_{kj} F_{kk} + F_{jk} + \delta_{jk}(\alpha_j^2 + F_{jj})\right) - i\Delta_j F_{jk} \\
&\quad - iJ(F_{jk+1} e^{-i\phi} + F_{jk-1} e^{i\phi}) - \frac{\kappa}{2} F_{jk} + (j \leftrightarrow k).
\end{aligned} \quad (3)$$

In Supplemental Material [49], we benchmark the Gaussian ansatz results with exact ones for a single site. Extending the comparison to larger systems is prohibitive due to the large size of the local Hilbert space. However, the Gaussian ansatz is expected to be valid as long as $\langle b_j^\dagger b_j \rangle / |\alpha_j|^2 \ll 1$. Indeed, in this limit, H_a can be approximated by a quadratic Hamiltonian in terms of fluctuation operators, such that the steady state is close to Gaussian. The above condition is met by our results; see Supplemental Material [49].

Stability diagram—We have numerically integrated Eq. (3) with a homogeneous profile for the driving amplitude $|\epsilon_j| = \epsilon$ and detuning $\Delta_j = \Delta$ and obtained that, in most of the phase diagram, the system exhibits persistent fluctuations at long times and fails to reach a steady state.

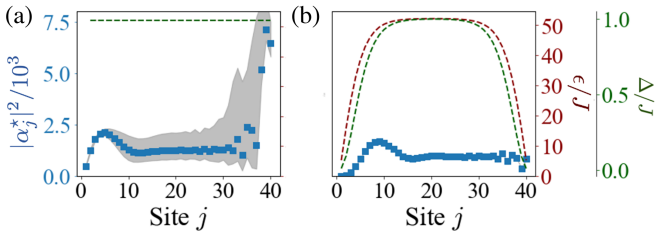


FIG. 1. Long-time limit of α_j obtained with (a) a homogeneous profile ($\epsilon_j = \epsilon$) and (b) the inhomogeneous profile in Eq. (4). The gray area in (a) refers to the range of values that α_j takes in the oscillating long-time dynamics. Dashed red and green lines refer to profiles of ϵ_j and Δ_j, respectively. $N = 40$, $\phi = \pi/3$, $\kappa/J = 1$, and $U/J = -2 \times 10^{-4}$.

Oscillatory solutions may indicate the presence of Liouvillian modes with a vanishing real part in the thermodynamic limit [50]. This happens even though $\kappa \neq 0$ in all our calculations. We have found that an inhomogeneous profile of the amplitude (ϵ_j) and local drive detuning Δ_j in which these quantities are suppressed toward the ends of the chain attenuates the effect of open boundaries and stabilizes the steady state [51]. Throughout the Letter, we choose, for $j \leq N/2$,

$$\epsilon_j = \epsilon \tanh(j/N_0), \qquad \Delta_j = \Delta \tanh^2(j/N_0), \tag{4}$$

and a mirror-reflected pattern for $j > N/2$. N_0 in Eq. (4) is the size of the border region. These smooth boundary conditions produce stable solutions in a larger region of the nonequilibrium phase diagram, such that $\alpha_j(t)$ converge toward a steady-state value, $\alpha_j(t \to \infty) = \alpha_j^\star$. Figure 1 shows a typical comparison between solutions with (a) homogeneous drive (i.e., sharp boundary conditions) and (b) the smoothened boundary conditions, showing the stabilization of the steady state (further examples in Supplemental Material [49]).

Nonequilibrium steady-state phase diagram—We focus on a regime of weak nonlinearities $|U| \ll \epsilon_j$, where the Gaussian ansatz is more accurate. This condition is also easily met in experimental implementations, where nonlinearities are typically small. Figures 2(a) and 2(b) show the calculated phase diagram. In Fig. 2(a), we recognize a low-density region phase (I). The rest of the phase diagram comprises a high-density region (III) and a coexistence region (II), a division that is not apparent in Fig. 2(a) but will become evident below. This phase diagram is in sheer contrast with the $\phi = 0$ case (see Supplemental Material [49]). Finally, the gray area is a chaotic region in which the long-time behavior of the Gaussian ansatz does not converge to a steady-state solution. Figure 2(b) shows that fluctuations are enhanced at the critical line separating I and II and remain large within region II, which is in contrast with the $\phi = 0$ case; see Supplemental Material [49]. Figures 2(c) and 2(d) show the evolution of steady-state

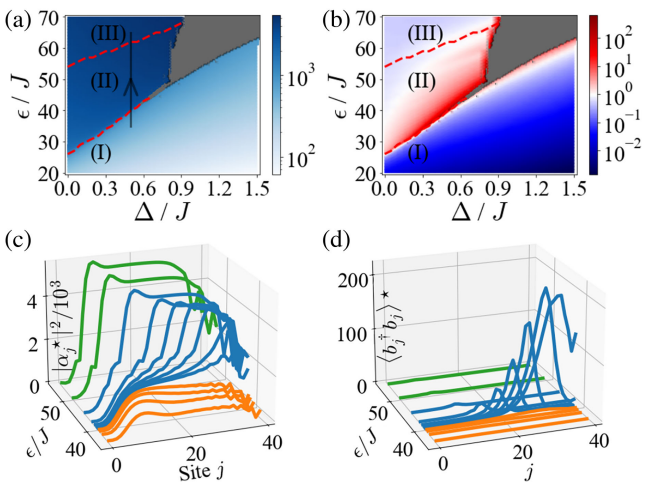


FIG. 2. Phase diagram of (a) mean displacement value $\overline{|\alpha^\star|^2} = (1/N)\sum_{j=1}^{N} |\alpha_j^\star|^2$ and (b) maximum value of fluctuations, $\langle b_j^\dagger b_j \rangle^\star$, in the chain. (c),(d) Steady-state profile of the chain $|\alpha_j^\star|^2$ and $\langle b_j^\dagger b_j \rangle^\star$ for phases I (orange, low density), II (blue, coexistence), and III (green, high density), whereas $N = 40$, $\Delta/J = 0.5$, $\phi = \pi/3$, $\kappa/J = 1$, and $U/J = -2 \times 10^{-4}$.

spatial profiles of the mean values and Gaussian fluctuations in the chain in the three regions of the phase diagram. Orange and green curves correspond to low-density (region I) and high-density (region III) phases. Coexistence region II (blue curves) shows a strong inhomogeneous photon density profile with large fluctuations. Below, we characterize quantitatively this phase in terms of non-Hermitian topology.

Topology in the phase coexistence region—The obtained nonequilibrium phase diagram can be understood under the lens of topological amplification theory. In the limit of weak nonlinearities, the Hamiltonian part of the dynamics of the b_j fields at long times is governed by the following effective Hamiltonian (to quadratic order in the $b_j, b_j^\dagger$ operators):

$$H_b \approx \sum_j \tilde{\Delta}_j b_j^\dagger b_j + \sum_{j=1}^{N} \left(g_j b_j^{\dagger 2} + g_j^* b_j^2\right) + J \sum_{j=1}^{N-1} \left(b_{j+1}^\dagger b_j e^{i\phi} + b_j^\dagger b_{j+1} e^{-i\phi}\right), \tag{5}$$

with the linear term vanishing from the $\langle b_j \rangle = 0$ condition, and $\tilde{\Delta}_j = \Delta_j + 4U|\alpha_j^\star|^2 + U$, $g_j = U(\alpha_j^\star)^2$. H_b describes fluctuations around the background provided by the coherent state component of the Gaussian ansatz. We emphasize that, even if U takes small values, the corrections to H_b are of the order of $U|\alpha_j|^2$ and can become comparable to J and κ.

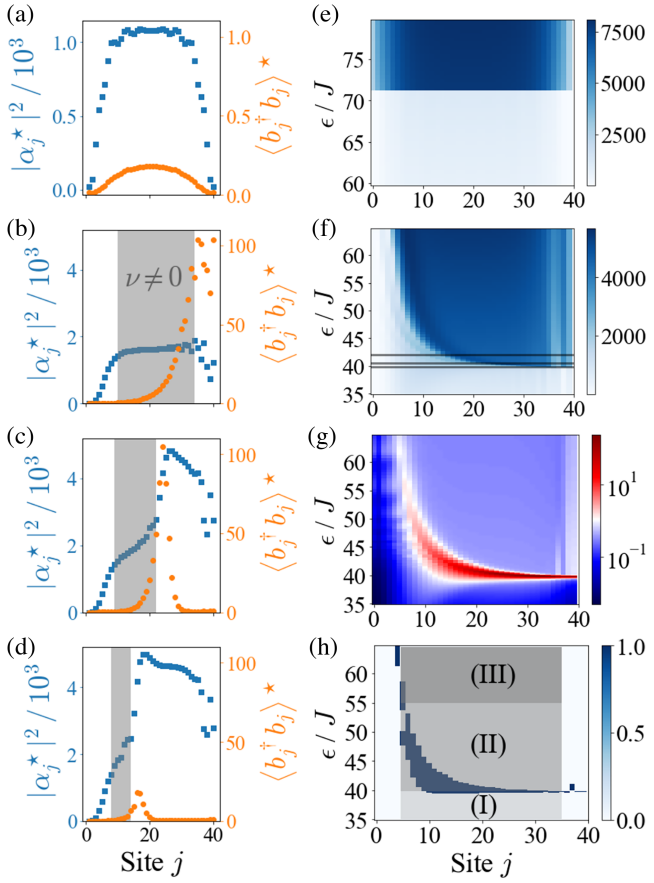


FIG. 3. Steady-state profile of the chain: (a) for $\phi = 0$, $\epsilon/J = 70$ and (b)–(d) for $\phi = \pi/3$ and three values of $\epsilon/J = 39.8, 40.5$, and 42. (e) Profile of $|\alpha_j^\star|^2$ when ϵ is varied for $\phi = 0$ and (f) profile of $|\alpha_j^\star|^2$, (g) fluctuations $\langle b_j^\dagger b_j \rangle^\star$, and (h) local winding number ν_j when ϵ is varied along a line of constant Δ/J on the phase diagram in Fig. 2(a), whereas $N = 40$, $\Delta/J = 0.5$, $\phi = \pi/3$, $\kappa/J = 1$, and $U/J = -2 \times 10^{-4}$.

To reveal the link with non-Hermitian physics, we define the set of $2N$ fluctuation operators in the Nambu basis, $\mathsf{b}_{\mu(=j)} = b_j$ (if $\mu = 1, \ldots, N$) and $\mathsf{b}_{\mu(=j+N)} = b_j^\dagger$ (if $\mu = N+1, \ldots, 2N$). In an input-output formalism [40,42], the linear response of these operators is $\dot{\mathsf{b}}_\mu = -i \sum_\nu \mathbb{H}_{\mu\nu} \mathsf{b}_\nu + F_\mu$, with F_μ representing input fields and

$$\mathbb{H} = \begin{pmatrix} D - i\frac{\kappa}{2}\mathbb{1} & K \\ -K^* & -D^* - i\frac{\kappa}{2}\mathbb{1}, \end{pmatrix}, \tag{6}$$

with $D_{jk} = \Delta_j \delta_{jk} + J(\delta_{j,k-1} e^{i\phi} + e^{-i\phi}\delta_{j,k+1})$ and $K_{jk} = 2g_j \delta_{jk}$. We characterize the topology properties of the steady state of our model (2) with the winding number

$$\nu = \frac{1}{2\pi} \mathrm{Im} \int_{-\pi}^{\pi} dk \mathrm{Tr} \partial_k \log[\mathbb{H}(k)], \tag{7}$$

where $\mathbb{H}(k)$ is the Fourier transform of the non-Hermitian matrix $\mathbb{H}$ and we assume an homogeneous system. In [42], it was shown that nonzero values of ν lead to the chain entering in a regime of directional amplification, which imply large elements of the zero-frequency Green's function, $\mathbb{G} = -\mathbb{H}^{-1}$. ν is also related to non-Hermitian point-gap topology [44] (see Supplemental Material [49] for details).

Let us focus now on the transition between low- and high-boson density phases in region II of the phase diagram. The coexistence region is bounded by values $\epsilon_{c,1}$ and $\epsilon_{c,2}$ of the coherent drive as we increase ϵ along lines of constant Δ. In Figs. 3(b)–3(d), we show the evolution of density profiles as we move along the line in Fig. 2(a). The chain is divided into two spatial regions of high- and low-boson densities, separated by an interface with large of fluctuations. The interface moves from right to left as we ramp up the value of ϵ. The topological trivial case $\phi = 0$, shown in Fig. 3(a) for comparison, does not show those spatial features. A topological characterization can be achieved by using the definition in Eq. (7). In a homogeneous system, $\mathbb{H}$ is parametrized by constant values $g_j = g$, J, and $\Delta_j = \Delta$ and the winding number is a function $\nu(g, J, \Delta)$. In order to characterize spatial regions in our inhomogeneous system, we define the function $\nu_j = \nu(g_j, J, \tilde{\Delta}_j)$ with the local Hamiltonian parameters. This local definition of a winding number is correct in a quasihomogeneous limit in which parameters vary slowly in space. We highlight that ν_j is implicitly dependent on ϵ_j, since it depends on $\alpha_j^\star$ through the relations below Eq. (5) and the steady-state value $\alpha_j^\star$ is determined by ϵ_j. Different values of ν_j across the chain imply coexistence of different topological phases, as we show below.

As we modify ϵ along a constant Δ line, we find (i) for $\epsilon < \epsilon_{c,1}$ (region I), a low-density phase with a spatially homogeneous density profile and $\nu_j = 0$. (ii) $\epsilon = \epsilon_{c,1}$ marks the onset of topological amplification [see Fig. 3(b)] and the transition between phases I and II. At this critical point, $\nu_j = 1$ for all j's except for the edges of the chain. At this point, corrections induced by interactions take significant values, $g_j/J \approx 0.4$, $(\tilde{\Delta}_j - \Delta_j)/J \approx 1.6$. (iii) $\epsilon_{c,2} > \epsilon > \epsilon_{c,1}$ is a phase coexistence region in which the chain is divided into a low-density, topological spatial region with $\nu_j = 1$ and a high-density topologically trivial region with $\nu_j = 0$; see Figs. 3(c) and 3(d). The topological and trivial spatial regions are separated by a moving edge with large fluctuations of the bosonic field. This numerical result agrees with previous works that predict the accumulation of bosons at the edge of nonreciprocal quantum chains [40,52]. The continuous evolution and phase coexistence are clearly seen in Fig 3(f), whereas such phase coexistence is absent in the topologically trivial $\phi = 0$ case; see Fig. 3(e). Figure 3(g) shows fluctuation maxima at the interface between topological and trivial phases. Finally, Fig. 3(h) shows the evolution of the topological region as identified by the

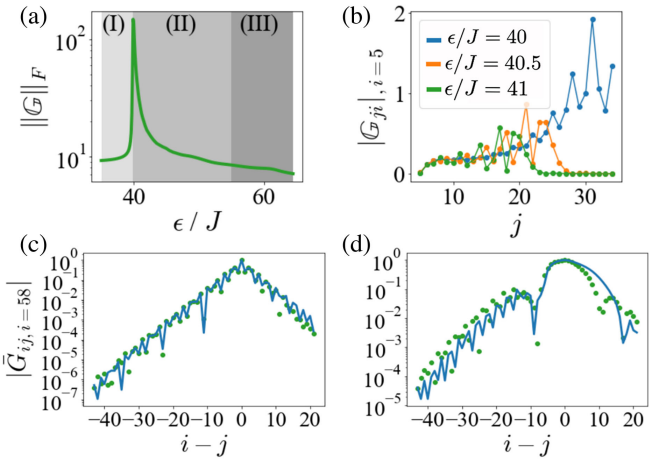


FIG. 4. (a) Norm of the Green's function $||\mathbb{G}||_F = \sqrt{\mathrm{Tr}(\mathbb{G}^\dagger \mathbb{G})}$ and (b) values of some nondiagonal elements of the Green function for $N = 40$, $\Delta/J = 0.5$, $\phi = \pi/3$, $\kappa/J = 1$, and $U/J = -2 \times 10^{-4}$. Normalized correlation function $\bar{G}_{jk}$ (c) $N = 80$, $\epsilon/J = 38$ (low-density phase), (d) $N = 80$, $\epsilon/J = 40.2$ (topological coexistence phase) for the nonlinear Bose-Hubbard chain (green dots), and the effective quadratic model extract from the steady-state value (blue line). $\Delta/J = 0.5$, $\phi = \pi/3$, $\kappa/J = 1$, and $U/J = -2 \times 10^{-4}$.

winding number ν_j. (iv) Finally, for values $\epsilon > \epsilon_{c,2}$ (region III), the whole chain enters into the high-density, non-topological phase. A quantitative definition of the boundary between II and III, $\epsilon_{c,2}$, is complicated by finite-size effects. In order to establish a criterion, we define $\epsilon_{c,2}$ as the point at which ν_j vanishes at all points j within the bulk of the chain, corresponding to the segment $N_0 < j < N - N_0$.

Green's function and correlation functions—Our calculations show that both the chain Green's function $\mathbb{G}$ and the correlations are enhanced at the topological phase. Topological properties of $\mathbb{G}$ can be derived by the theory of topological amplification (see [11,42] and Supplemental Material [49]). This is indeed what we observe in Fig. 4(a). Figure 4(b) also shows that, in the topological region of the chain, for a site i on the left-hand side of the chain, $\mathbb{G}_{ji}$ grows with j, which is a signature of directional amplification. We have also studied normalized two-point correlation functions $\bar{G}_{jk} = G_{jk}/\sqrt{n_j n_k}$ at the low-density phase and at the critical line separating phases I and II for a chain of $N = 80$ sites. In Figs. 4(c) and 4(d), we observe longer-range correlations on the critical line, whereas in the low-density phase, correlations decay exponentially with distance.

Conclusions and outlook—We have shown that a driven-dissipative Bose-Hubbard model with a site-dependent, linearly varying phase in the coherent drive exhibits topological amplification at the phase boundary between a high- and a low-density phase. Topological phases can be characterized by a topological invariant associated to point-gap topology. Our model shows the emergence of non-Hermitian topology and its physical consequences in a quantum *nonlinear* dissipative system.

Our model can be implemented with arrays of superconducting cavities, following the scheme in Ref. [51], in which the Kerr nonlinearity U arises naturally from Josephson junctions and dissipation κ from the coupling to external transmission lines. In addition, the phase gradient ϕ and the inhomogeneous profiles of amplitude ϵ_j and detunings Δ_j are induced via the external microwave pump and a local tuning of the cavity frequencies. Our Letter may lead to the design of nonlinear amplifiers and photodetectors in the microwave regime. Alternative topological invariants like the one introduced for a single bosonic mode in [53] may be compared to our model. Future work will involve characterizing critical properties and timescales, which may reveal anomalous large relaxation times [54]. It would be worth studying strongly interacting regimes with other numerical methods such as matrix product state methods [55,56], the appearance of similar phenomena in spin chains [57] and strongly interacting lattice models [58], and extensions to two dimensions where topological amplification can occur in chiral edge states [59].

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